

# Testing the make-them-run theory in elite badminton: spatial pressure and rally outcomes in women's singles

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A standard piece of badminton coaching advice is that rallies are won by forcing the opponent to cover more court. I test this “make them run” theory using ShuttleSet22, a stroke-level dataset of elite singles matches, focusing on the 13 women's-singles players in the dataset and roughly 22,500 strokes. I operationalize spatial pressure as displacement: the distance a shot forces the receiving player to move. Across three framings, the theory does not hold. Cumulative displacement applied over a rally is nearly identical for eventual winners and losers; data-driven playstyle clusters are weak among these elite players (silhouette  $\approx 0.22$ ) and absent for men; and the winner's penultimate “setup” shot is no larger than their own rally-wide average. What does distinguish rallies is the final shot: about 63% of rallies end on the losing player's racket as an error, and final-shot displacement tracks how the rally ended rather than who won, highest when the loser sends the shuttle out and lowest when they net it. The findings are descriptive and scoped to one spatial channel; they suggest that among elite players, displacement measures how frantic a rally became rather than who controlled it.

sports analytics | badminton | spatial data | clustering | exploratory data analysis

Ask a badminton coach how you win a long rally and you will hear some version of the same advice: move your opponent. Stretch them corner to corner, make them cover the court, and sooner or later they break down. The idea treats spatial pressure—how far and how often you force an opponent to travel—as the way to wins points, and at the club level it is hard to argue with. Whether it survives at the elite level, where every player defends well enough that a step out of position is rarely fatal, is a question stroke-level tracking data can now test.

## Significance

Badminton coaching wisdom holds that you win rallies by moving your opponent around the court. Using stroke-level tracking data from elite women's singles, this study tests that idea directly. It finds that the amount a player moves their opponent (spatial pressure) does not separate rally winners from losers: players apply nearly identical pressure throughout a rally, distinct playstyles do not emerge cleanly among elite players, and winners do not engineer the finish with one decisive setup shot. Instead, rallies are settled on a single final shot, most often the trailing player's error. The result reframes a common coaching heuristic and shows the limits of one widely tracked spatial metric.

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Recent work has built unusually detailed shot-by-shot records of professional play. ShuttleSet22 (1) records, for every stroke, where both players stood, where the shuttle landed, and the shot type; on the final stroke of a rally it adds who won and how the point ended. It extends an earlier annotated dataset, ShuttleSet (2), both of which grew out of the wider CoachAI effort to collect fine-grained badminton data (3). The data was built for one job in particular: *stroke forecasting*, which predicts a player's next shot based on the rally so far (4). I want to ask something the same data can answer but was not built for: does spatial pressure, which refers to how far a player moves their opponent, decide who takes the rally? Put that way, the project is less a model than a direct test of a belief that almost everyone in the sport already takes for granted, and it sits alongside a longer line of work that uses spatial tracking to study tactics in team sports like basketball and soccer (5).

I treat spatial pressure as one concrete quantity, *displacement*: the straight-line distance from where the receiving player is standing to where the shuttle lands, in the dataset's own court-coordinate units. A large displacement means the shot dragged the opponent a long way; a small one means it landed close to them. This is deliberately one narrow channel of pressure. It says nothing about pace, deception, or shot quality—any of which could carry a signal that displacement misses—so I scope every claim below to the spatial channel alone. I keep the main analysis on women's singles, where the data supports a weak playstyle typology; the men's draw does not cluster at all, which I treat as a finding about how uniform elite men are rather than as a flaw in the method.

I test the make-them-run theory in three ways, and each time it comes up short. The cumulative displacement a player piles up over a rally does not separate eventual winners from losers. Players do not fall into displacement-based styles that win in different ways. And the winner's setup shot, which is the contact right before the rally ends, is no more displacing than that player's ordinary shot. What is left is the final shot, as a rally is usually decided by one contact, and that contact is more often the trailing player's error than the leader's clean winner. The displacement on that last shot reflects *how* the rally ended, not who deserved to win it.

## Data

The data comes from ShuttleSet22 (1), a public stroke-level dataset of elite professional singles matches released to support stroke-forecasting research. The full set spans 58 matches and 35 players recorded between 2020 and 2022, and it contains 52,356 strokes across 4,944 rallies. These counts reflect the data as it is publicly distributed, which is larger and spans more seasons than the benchmark splits described in the source paper.

In the original source, the unit of analysis is the individual stroke. Each stroke record carries the shot type (labeled in Chinese, which has translations to English), the court positions of both players, where the shuttle landed, and—on the final stroke of a rally—the outcome and the reason the rally ended. I keep the stroke as my unit for most of the analysis; for the playstyle clustering, I roll strokes up to the player level, and for the pressure and outcome analyses, up to the rally-player level (one row per player per rally).

I restrict the main analysis to women's singles. The data does not record player sex, so I infer it from who plays whom: matches are never mixed, so the two players in any match share a sex, and I propagate a handful of known-sex labels outward through the graph of shared matches. That gives 22 men and 13 women, and the women's-singles subset comes to roughly 22,500 strokes.

Two properties of the source limit what the numbers can say. First, court positions and landing spots come in the dataset's own camera-derived coordinate system, not in meters. The dataset includes a homography to real-world coordinates—which I do not apply—so every displacement is relative. I can say one shot moved a player about twice as far as another, but I cannot put a metric distance on it. Second, strokes per player are uneven (a few hundred for some, several thousand for others), so the per-player feature estimates are noisier for low-volume players, and I flag that where it touches the clustering.

The missingness sits in a few fields. Serves and a handful of other strokes have no recorded opponent position, so they have no displacement either; I drop them from the displacement analyses but keep them in the counts. The rally winner appears only on the last stroke of each rally, so I propagate it back to every stroke in that rally (described in Methods). Where a stroke is missing a rally outcome or a displacement value, I leave it out of the analysis that needs the field, and I flag those drops in the relevant figures.

## Methods

The analysis is descriptive and exploratory throughout. I make no causal claims and use no train-test split, because the aim is to describe what co-varies with rally outcomes, not to build a predictor or pin down a cause. The pipeline runs as a sequence of numbered notebooks (01–11) in the repository, and I point to the relevant one at each step.

**Cleaning and feature construction.** Notebook 02 concatenates the per-set stroke files from all 58 matches into one long table of 52,356 strokes and tags each stroke with its match, set, and rally identifiers. I map the 18 Chinese shot-type labels to English with a fixed dictionary. For every stroke I then compute the main derived variable, *displacement*, as the Euclidean distance between the opponent's recorded position and the shuttle's landing spot:

$$\text{displacement} = \sqrt{(x_{\text{opp}} - x_{\text{land}})^2 + (y_{\text{opp}} - y_{\text{land}})^2}.$$

Displacement is in the dataset's court-coordinate units and stays a relative measure throughout.

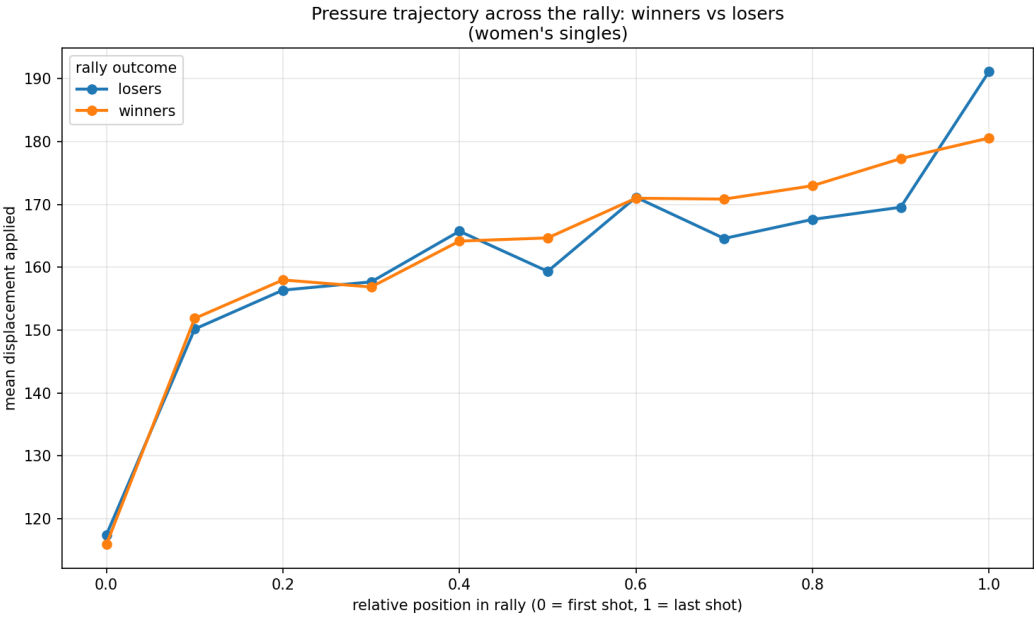
**Attaching rally outcomes.** The winner is recorded only on the final stroke of each rally, in the `getpoint_player` field. To label every stroke with its rally's eventual winner, I group strokes by match, set, rally and forward-fill and back-fill that field within each group. This is a within-group fill, not a table join, so it leaves the row count unchanged; notebook 02 prints stroke counts before and after to confirm. From the filled field I build `hitter_won_rally`, a Boolean that is true when the player who hit a stroke went on to win the rally.

**Resolving player identity.** The stroke tables do not carry player names directly; they encode players as "A" (match winner) or "B" (match loser). I resolve these to real names with a left join of each stroke onto the match-metadata table on the match identifier, then map A and B to the winner and loser names. The join is one-to-one on match identity and does not change the stroke count; the shared helper that performs it (used by notebooks 04, 06–09, and 11) prints row counts before and after and asserts the count is unchanged, which guards against an accidental fan-out.

**Variables.** The main variables are: *displacement* (per stroke, defined above); *cumulative displacement*, the running sum of a player's displacement within a rally; *hitter\_won\_rally* (per stroke); *from\_end*, a stroke's position counted backward from the rally's last shot (0 = final shot, 1 = penultimate or "setup" shot); and *win\_reason*, the recorded cause of the rally ending (a clean placement winner versus a forced opponent error of a given type). For the clustering, I summarize each player with a feature vector: shot-type mix, mean displacement applied, court depth, rally lengths, and the share of their rally wins that end on their own winner rather than an opponent error.

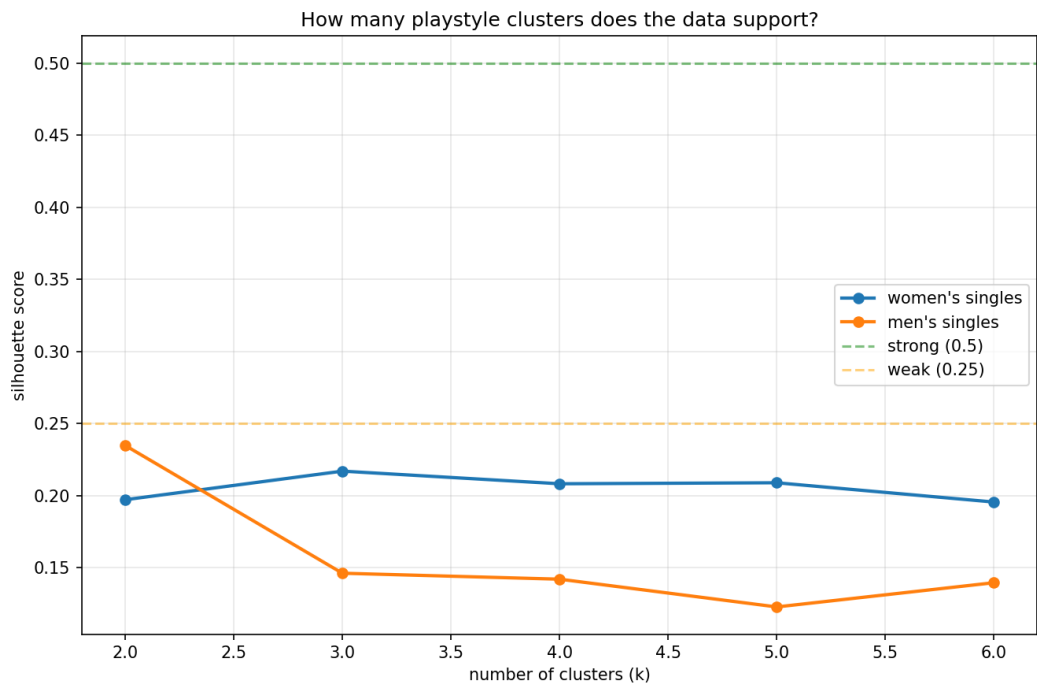
**Analyses.** Five analyses follow. (i) A descriptive comparison of cumulative displacement over the rally for eventual winners versus losers (notebook 03). (ii) A data-driven playstyle typology: I standardize the per-player features, cluster them with hierarchical (Ward-linkage) agglomerative clustering, project the result to two dimensions with PCA for visualization only, and pick the number of clusters with silhouette scores across  $k = 2$  to 6 (notebooks 04–05b). (iii) Logistic regressions of rally win on standardized displacement, fit separately within each women's playstyle cluster (notebook 06); these are associational and, given the weak clusters, I read them as suggestive only. (iv) A comparison of final-stroke displacement across rally-ending types (notebook 08). (v) A test of whether the winner's setup shot is unusually displacing relative to their own rally-wide baseline, plus a per-position trajectory of displacement over the closing shots (notebooks 09 and 11). Throughout, I lean on effect sizes relative to the shot-to-shot standard deviation of displacement rather than formal significance tests, given the exploratory aim.

**Cumulative pressure does not separate winners from losers.** Follow the cumulative displacement a player piles up over a rally and the eventual winners and losers look nearly identical at every rally length (Figure 1). Across all strokes, eventual winners averaged 155.5 units of displacement and eventual losers 156.4, a gap that vanishes against ordinary shot-to-shot variation (the stroke-level standard deviation is about 78). So the make-them-run idea, in its simplest cumulative form, does not pick out who wins. Table 1 summarizes the analysis sample.



**Fig. 1.** Mean displacement applied by the hitter at each relative position in the rally (0 = first shot, 1 = last shot), split by whether the hitter went on to win. The two curves track each other closely the whole way, so eventual winners do not apply systematically more spatial pressure than eventual losers. Women's singles; displacement in court-coordinate units.

**Elite players do not separate into clean playstyles.** A data-driven typology does not rescue the theory either. Clustering all 35 players at once splits men from women rather than tactical styles—a divide that comes from physical differences between the draws—so I restrict to one sex to clear that confound. Within women’s singles the clustering gives a weak three-group structure that peaks at a silhouette of about 0.22 at  $k = 3$ , under the conventional 0.25 line for even “weak but real” separation (Figure 2). Within men’s singles no  $k$  produces interpretable structure at all. Logistic regressions of rally win on displacement, fit within each women’s cluster, return mostly null associations, and the one nominally non-null cell rests on a pair of players, so I do not read into it. Among these elite players, shot composition and spatial-pressure profiles are too alike for a style-based edge to show up in this metric.



**Fig. 2.** Silhouette scores for hierarchical clustering of players across candidate cluster counts  $k$ , computed separately for women’s and men’s singles. Higher is better; the dashed lines mark the conventional thresholds for weak (0.25) and strong (0.5) separation. Women’s singles peaks at roughly 0.22 near  $k = 3$ , just under the weak threshold, while men’s singles never clears the noise floor. Neither draw supports a clean playstyle typology.

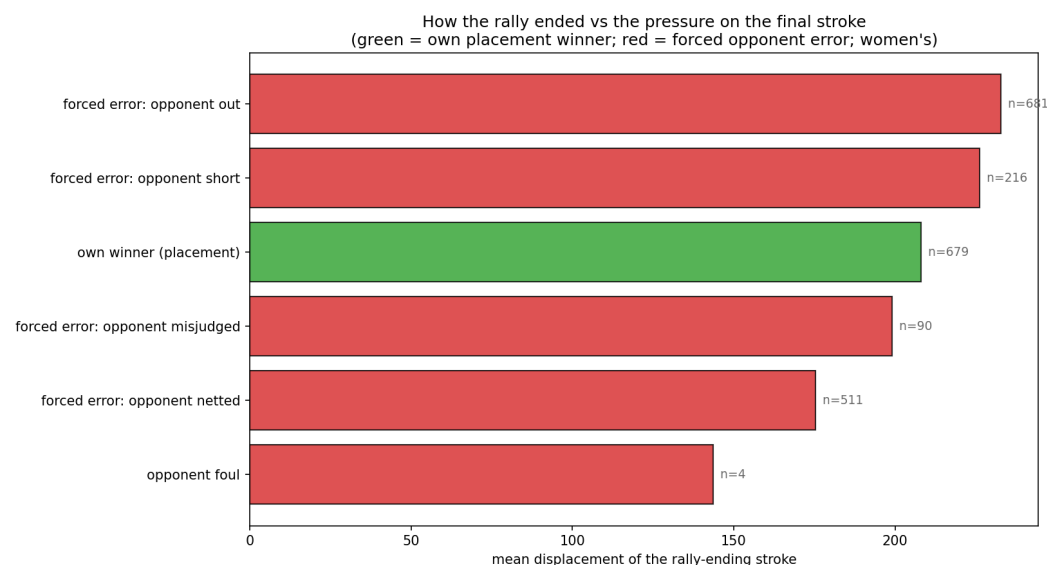
**The rally is decided on the final shot.** What does separate rallies is how they end. About 63% of rallies end on the losing player’s racket. The point closes because the trailing player misses—nets the shuttle or sends it out—not because the leader hits an untouchable winner. Final-stroke displacement then tracks the *type* of ending rather than who won (Figure 3): it is highest when the opponent sends the shuttle out of bounds (mean 233), next when they fall short or get beaten by a placement, and lowest when they net it (mean 175). The netted bar is the cleanest evidence that this is not just geometry. A netted shuttle lands short, right near the court, yet it still shows the lowest displacement of all, which is what you would expect from errors made under almost no spatial pressure.

**Winners do not engineer the finish.** Finally, the winner does not manufacture the finish with one decisive setup shot. The winner’s penultimate (“setup”) shot averages 163 units of displacement, essentially the same as that winner’s own rally-wide baseline of 159, a gap of under four units. Over the closing shots the winner and loser curves weave around each other within noise until the final contact, where both spike and the loser’s last shot is actually the larger of the two (Figure 4). The decisive gap sits entirely in the last shot, and it is the loser overhitting rather than the winner placing.

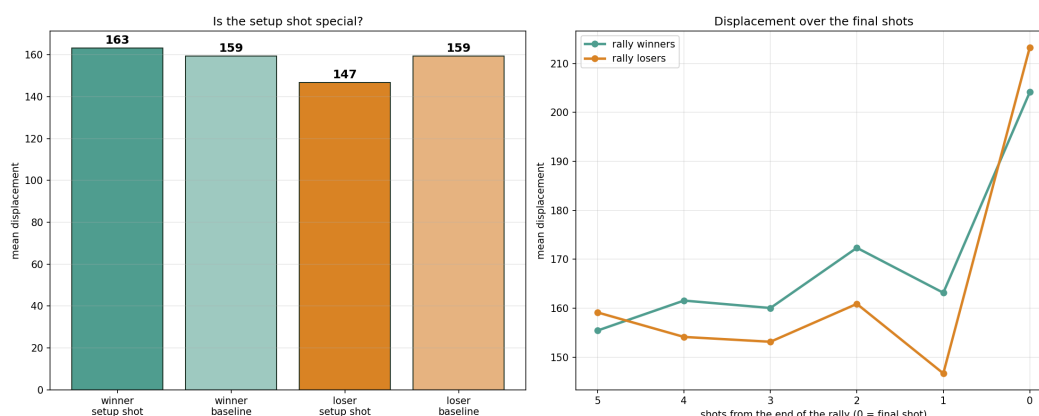
**Discussion**

Across three framings, the make-them-run theory does not account for who wins an elite women’s-singles rally, at least not in any way displacement can detect. Players apply nearly the same cumulative spatial pressure whether they win or lose; they do not split into displacement-based styles that win in different ways; and winners do not set up the finish with an unusually displacing shot. What does separate rallies is the final contact, and that contact is more often the trailing player’s miss than the leader’s clean winner. Final-shot displacement reflects how the rally ended, not who controlled it, so among these players displacement seems to track how frantic a rally got rather than who was winning it.

These findings come with clear limits. The analysis is descriptive and associational, not causal: it shows what co-varies with rally outcomes, not what causes them. Displacement also captures only one channel of pressure—the spatial “make them run” dimension—and says nothing about pace, deception, timing, or shot quality, any of which



**Fig. 3.** Mean displacement of the rally-ending stroke, grouped by how the rally ended. Forced opponent errors (red) span the range, with out-of-bounds errors the highest and netted errors the lowest; an own placement winner (green) sits in between. Netted errors show the lowest displacement despite landing short near the court, so the pattern is not a geometric artifact of where out-of-bounds shuttles land. Women's singles; bar labels give sample sizes.



**Fig. 4.** The winner does not engineer the finish. Left: mean displacement of the winner's and loser's setup (penultimate) shots against each player's own rally-wide baseline; the winner's setup shot (163) is essentially equal to their baseline (159). Right: displacement over the closing shots of the rally (0 = last shot), where the two curves stay close until the final contact, then both rise and the loser's last shot is the larger. Women's singles; displacement in court-coordinate units.

**Table 1. Characteristics of the data and the women's-singles analysis sample. Rows marked "full dataset" cover all 58 matches; the rest describe the 13-player women's-singles subset I analyze, after dropping strokes with no recorded displacement or rally outcome where a field is required. Displacement is in the dataset's court-coordinate units.**

| Quantity                                      | Value     |
|-----------------------------------------------|-----------|
| Matches (full dataset)                        | 58        |
| Players (full dataset)                        | 35        |
| Strokes (full dataset)                        | 52,356    |
| Rallies (full dataset)                        | 4,944     |
| Women's-singles players                       | 13        |
| Women's-singles strokes                       | ≈22,500   |
| Time window                                   | 2020–2022 |
| Mean displacement, all strokes (full dataset) | 156.0     |
| Mean displacement, winners (full dataset)     | 155.5     |
| Mean displacement, losers (full dataset)      | 156.4     |
| Stroke-level SD of displacement               | ≈78       |
| Rallies ending on the loser's racket          | 63%       |

could carry the signal this metric misses. The coordinates are relative rather than metric, the women's-singles sample is small (13 players), and the clusters are weak enough that I read the style-conditional results as suggestive only. Future work could apply the dataset's homography to recover real distances, fold in the hitter's own court position to tell controlled pressure apart from scrambling, and push the analysis below the elite level, where players may separate into styles more cleanly than the top of the tour does.

## Agentic Analysis

I used Claude on this project, both for the analysis code and the writing. This section covers how I worked with it: what I asked, what I kept, what I overrode, and where it got things wrong.

**How I worked.** Going into the project, I did not want to include anything I did not check. I had the framing fixed before I opened a session: scope every claim to spatial displacement, report effect sizes instead of significance, and keep the analysis direction fairly descriptive. I handed Claude that framing, the structure of the dataset, and the hypothesis, and let it work inside those limits. Then I checked what it gave back against the real numbers instead of taking its word for what it had produced (something I was worried it would get wrong often). The analytical and interpretive calls were mine; Claude drafted and ran code inside them.

**What I accepted.** I took suggestions when I could see for myself why they were right, be it checking the data on my own or using my own judgment to confirm the logic. On the code side that meant pulling repeated data-prep logic into one shared helper module with its own row-count diagnostics, and a bootstrap step that lets the code find the data no matter which directory I run it from. On the writing, I kept the suggestions that held the scope I had already set and dropped the ones that wandered off it.

**What I pushed back on.** I pushed back on the claims. If left alone, the analysis leaned toward a cleaner, louder claim: that spatial pressure just does not matter. I held it to the version the data supports, where the winner-loser gap sits around 0.2 standard deviations, inside the noise, and goes in as an effect size rather than a significant result. When one regression cell came back nominally non-null, I left it out instead of reporting it, since it rested on just two players. I could hold that line because of my badminton knowledge, the model did not have: elite players are not recreational players, and at the top of the sport everyone defends well, so finding no spatial-pressure edge is a statement about how even the elite field is, not about badminton as a whole. That is why the results are scoped and hedged the way they are.

**Where the assistant went wrong.** Its worst habit was overstatement. The default was to write findings as bigger and more certain than they were, to round a hedged, modest result up into a clean headline. On a paper whose main results are null or close to it, that is the failure that matters most, and pulling claims back to size was a constant job rather than a one-time edit.

It was also least trustworthy when describing its own output. At one point it read its own setup-shot figure wrong and told me the winner's setup shot stood out, when the chart had it sitting right on the player's baseline; another time it gave me a summary statistic that turned out to be computed wrong and had to be redone from the data. That is the whole reason I checked every number against the source rather than trusting its account of what it had made.

**Transcript Location.** The full session transcripts are in the project repository.

**Data, Materials, and Software Availability.** All data, analysis code, and the notebooks that produce every figure and table are available at <https://github.com/aakankshvaidya27/qss20-final-project>. The source dataset, ShuttleSet22, is publicly available (1).

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